Experiment 6: Time Series Analysis of Website Traffic

Objective: To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
# Create website traffic data
data = {
    "Date": [
        "2026-03-01",
        "2026-03-02",
        "2026-03-03",
        "2026-03-04",
        "2026-03-05",
        "2026-03-06",
        "2026-03-07"
    ],
    "Visitors": [150, 210, 175, 290, 230, 340, 300]
}
# Create DataFrame
df = pd.DataFrame(data)
# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])
# Set Date as index
df.set_index("Date", inplace=True)
# Display the data
print("Website Traffic Data:")
print(df)
```

```
# Display basic information
print("\nStatistical Summary:")
print(df.describe())
# Calculate average visitors
print("\nAverage Website Visitors:", df["Visitors"].mean())
# Find maximum visitors
print("Maximum Website Visitors:", df["Visitors"].max())
# Find minimum visitors
print("Minimum Website Visitors:", df["Visitors"].min())
# Plot the time series
plt.figure(figsize=(10, 5))
plt.plot(
    df.index,
    df["Visitors"],
    marker="o"
)
plt.title("Time Series Analysis of Website Traffic")
plt.xlabel("Date")
plt.ylabel("Number of Visitors")
plt.xticks(rotation=45)
plt.grid(False)
plt.tight_layout()
plt.show()
```

OUTPUT:

```
Website Traffic Data:
            Visitors
Date
2026-03-01       150
2026-03-02       210
2026-03-03       175
2026-03-04       290
2026-03-05       230
2026-03-06       340
2026-03-07       300

Statistical Summary:
         Visitors
count    7.000000
mean   242.142857
std     69.991496
min    150.000000
25%    192.500000
50%    230.000000
75%    295.000000
max    340.000000

Average Website Visitors: 242.14285714285714
Maximum Website Visitors: 340
Minimum Website Visitors: 150
```

